Aranyabodh Annual Operating Cost Projection

November 2026 to October 2027

Prepared 15 October 2026 | Currency INR | Planning population 1,000 registered users

Executive assessment

Aranyabodh should reserve INR 8.60 lakh for the next 12 full operating months. The base projection totals INR 7.26 lakh in recurring cash requirements, including a provisional tax allowance. Adding 15% contingency and a separate INR 10,000 initial onboarding allowance produces a calculated funding requirement of INR 8.45 lakh. The rounded funding recommendation provides a further margin.

Annual funding measure	INR
Recurring services and operating allowances before tax	615,449
Provisional tax allowance at 18%	110,781
Recurring cash requirement including tax allowance	726,230
Contingency at 15% of recurring cash requirement	108,935
Initial onboarding envelope separate from recurring costs	10,000
Calculated first year funding requirement	845,165
Recommended first year funding allocation	860,000

Funding schedule: INR 50,000 per month in November and December 2026, followed by INR 75,000 per month from January to October 2027. Add INR 10,000 in November for initial provisioning and document ingestion. The annual allocation is therefore INR 860,000.

Purpose and cost boundary

This report supports operating-budget approval for a forest-document question answering service used by forest department officers and commercial entities. It covers the app backend, document processing, database and retrieval, model APIs, storage, backups, communications, monitoring and modest business-tool subscriptions.

Development, maintenance labour, salaries, manual document curation and sales labour are excluded. Subscription revenue is not offset against costs. Revenue-dependent payment processing and procurement-specific requirements are shown separately because no collection volume or procurement terms have been established.

All figures are estimates. Published provider rates, project evidence and planning assumptions are identified separately. Contingency and optional-service provisions are funding allowances rather than confirmed invoice charges.

1 Planning assumptions

The model holds the registered population and average activity constant for the full year. This gives a steady operating baseline rather than an acquisition or adoption forecast. Calendar-day calculations capture the variation between months.

Assumption	Base case and evidence status
Registered population	1,000 users. User-specified total, not 1,000 daily users.
Monthly active users	600 users or 60% of registered users. Planning assumption.
Daily active users	200 users or 20% of registered users. Planning assumption.
DAU to MAU ratio	33.3%. Equivalent to about 10 active days per MAU in a 30-day month.
Questions per active day	4.5 per DAU. Midpoint of the user-specified range of 4 to 5.
Monthly question volume	DAU × 4.5 × actual calendar days. 27,000 in a 30-day month.
Annual question volume	328,500 questions across 365 days.
Question mix	25% simple, 50% procedural and 25% complex. Assumed.
Initial corpus	500 documents × 20 pages = 10,000 pages. Assumed.
Monthly additions	100 documents × 20 pages = 2,000 pages. Assumed.
OCR share	50% of pages require OCR. Native-text files avoid OCR. Assumed.
Planning exchange rate	INR 90 per USD. Budget convention, not a verified spot rate.

Why these assumptions are usable

A specialist reference app is likely to be used when an officer or business needs to interpret a document or complete a procedure. The base case therefore assumes intermittent use by much of the population and repeated use by a smaller working group. It is not supported by a measured industry benchmark; the low and high scenarios test that uncertainty explicitly.

The repo describes a Gemini-backed retrieval service using Supabase Postgres with pgvector, document ingestion and citation-based answers. This architecture supports the selected cost categories. Current source files do not establish the live production capacity, enabled integrations or actual user behaviour. [10]

Tax and exchange rate treatment

An 18% tax cash allowance is applied to the complete operating subtotal for conservative budgeting. This is not an invoice-specific GST determination. Recoverable input credits, reverse-charge treatment and supplier differences are not netted off. The 15% contingency covers moderate currency movement and spending uncertainty; no separate foreign-exchange surcharge is added.

2 AI usage and unit economics

One user question can trigger several model calls for rewriting, planning, answering and verification. The following per-question allowances aggregate those generation stages. They do not represent the price of one API call. Thinking tokens are included in billable output. [1, 10]

USD per million tokens	Input	Output
Gemini 3.5 Flash Lite throughout projection	0.30	2.50
Gemini 3.8 Flash through December 2026	0.75	3.75
Gemini 3.8 Flash from January 2027	1.50	7.50

Google publishes the Flash increase effective 1 January 2027. The projection retains that announced change and holds all other model rates constant. It assumes standard paid API usage without caching discounts, batch discounts or promotional credits. [1]

Question class	Share	2026 INR	Flash bill share	2027 INR
Simple	25%	0.100	0%	0.100
Procedural	50%	0.600	80%	1.080
Complex	25%	1.500	90%	2.850

Class costs and Flash bill shares are planning assumptions. The repo records selected procedural generation costs of approximately INR 0.296, 0.764 and 0.682, while other runs show substantially higher costs when reasoning or retries increase. These examples support a broad cost range but do not validate the assumed mix or a fixed maximum. [10]

Reproducible calculation

2026 weighted generation cost = (25% × 0.10) + (50% × 0.60) + (25% × 1.50) = INR 0.70 per question.

Add 10% for recovery attempts: INR 0.70 × 1.10 = INR 0.77000 per question.

2027 class cost = 2026 class cost × (1 + assumed Flash share of its bill). The resulting weighted cost is INR 1.27750; including recovery attempts gives INR 1.40525 per question.

For a 30-day base month: 200 × 4.5 × 30 = 27,000 questions. Generation is INR 20,790 through December and INR 37,941.75 from January. The 10% recovery allowance is already included and is not applied again elsewhere.

Other API consumption

The INR 500 monthly OCR and embedding envelope assumes 1,000 scanned pages, 1.44 million document tokens after chunk overlap and 200 query-embedding tokens per question. At 27,000 questions, indicative costs are INR 135 OCR, INR 26 document embeddings and INR 97 query embeddings, leaving headroom. OCR free allowances are conservatively ignored; advanced extraction is not included. [1, 4]

3 Detailed recurring cost breakdown

All amounts below are INR before the provisional tax allowance. The two monthly columns use 30 days for comparison. The annual column uses 365 calendar days. Allowances are retained in the funded base case even if actual invoices are lower.

Cost category	2026 month	2027 month	Year total
AI generation	20,790	37,942	426,749
Backend API hosting	2,250	2,250	27,000
Document processing worker	2,250	2,250	27,000
Supabase Pro and Small compute	2,700	2,700	32,400
Redis or managed job queue	900	900	10,800
Storage backup and bandwidth envelope	1,000	1,000	12,000
Monitoring logs and error tracking	1,000	1,000	12,000
OCR and embedding envelope	500	500	6,000
Email SMS and WhatsApp envelope	2,000	2,000	24,000
Domain annual fee allocation	125	125	1,500
Business and support subscriptions	2,000	2,000	24,000
Optional Jev provision	1,000	1,000	12,000
Static frontend DNS CDN and TLS	0	0	0
Subtotal before tax	36,515	53,667	615,449

Hosting: one API service and one worker at USD 25 each per month. Redis is provisioned at USD 10. These are initial capacity allowances based on Render service examples; they are not a throughput guarantee or a high-availability deployment. [2]

Database: USD 25 Pro + USD 15 Small compute − USD 10 compute credit = USD 30 monthly. The base stays within 8 GB database disk and 250 GB uncached egress per month. Corpus growth, chat retention or retrieval traffic can exceed those quotas. [3]

Storage: assume 50 GB of originals and 50 GB of backup copies. At R2 standard rates, storage itself is approximately INR 122 monthly after a 10 GB free allowance. The rest of the INR 1,000 envelope covers operations, traffic elsewhere and backup tooling. It does not buy a managed point-in-time recovery service. [5]

Subscriptions: INR 1,000 monitoring, INR 2,000 communications and INR 2,000 business tools are provisional caps, not confirmed vendor subscriptions. Communications allow roughly 1,000 SMS, 4,000 WhatsApp utility messages, 6,000 emails and the WhatsApp platform fee. Marketing messages are excluded. [6]

Optional Jev: the repo defaults its modes to off. The provision covers about INR 817 at 27,000 questions if aggregate decision input averages 8,000 tokens per question at USD 0.042 per million. Its inclusion is conservative and assumes no reduction in Gemini cost. [7, 10]

4 Monthly operating cash projection

The projection covers 1 November 2026 through 31 October 2027. Other costs include all non-generation categories on page 4, totalling INR 15,725 monthly. Cash need includes the provisional tax allowance and excludes contingency and initial onboarding.

Month	Days	Questions	AI cost	Other costs	Tax	Cash need
Nov 2026	30	27,000	20,790	15,725	6,573	43,088
Dec 2026	31	27,900	21,483	15,725	6,697	43,905
Jan 2027	31	27,900	39,206	15,725	9,888	64,819
Feb 2027	28	25,200	35,412	15,725	9,205	60,342
Mar 2027	31	27,900	39,206	15,725	9,888	64,819
Apr 2027	30	27,000	37,942	15,725	9,660	63,327
May 2027	31	27,900	39,206	15,725	9,888	64,819
Jun 2027	30	27,000	37,942	15,725	9,660	63,327
Jul 2027	31	27,900	39,206	15,725	9,888	64,819
Aug 2027	31	27,900	39,206	15,725	9,888	64,819
Sep 2027	30	27,000	37,942	15,725	9,660	63,327
Oct 2027	31	27,900	39,206	15,725	9,888	64,819
Year	365	328,500	426,749	188,700	110,781	726,230

How to read the monthly movement

November to December changes reflect calendar length. The step in January primarily reflects the announced Flash price increase. February is lower because it contains 28 days. Later changes reflect calendar length; no price inflation or user growth has been assumed.

Generation cost = question volume × applicable per-question rate. Other costs = INR 15,725. Tax allowance = 18% × (generation + other costs). Monthly cash need = generation + other costs + tax allowance.

Recurring spend composition

Annual measure	INR
AI generation before tax	426,749
All other recurring services and allowances before tax	188,700
Provisional tax allowance	110,781
Average monthly cash requirement	60,519
Cash requirement per registered user per month	61

Amounts are calculated at full precision and displayed to the nearest rupee. Displayed rows can differ from their displayed total by a few rupees because totals are rounded only after summation. Service subscriptions and annual renewals may also have different invoice dates from these monthly allocations.

5 Funding schedule and initial costs

Contingency is 15% of recurring cash need. The recommended allocation rounds the monthly funding requirement to INR 50,000 through December and INR 75,000 thereafter. An additional INR 10,000 envelope is assigned to November.

Month	Cash need	Contingency	Allocation	Cumulative allocation
Nov 2026	43,088	6,463	60,000	60,000
Dec 2026	43,905	6,586	50,000	110,000
Jan 2027	64,819	9,723	75,000	185,000
Feb 2027	60,342	9,051	75,000	260,000
Mar 2027	64,819	9,723	75,000	335,000
Apr 2027	63,327	9,499	75,000	410,000
May 2027	64,819	9,723	75,000	485,000
Jun 2027	63,327	9,499	75,000	560,000
Jul 2027	64,819	9,723	75,000	635,000
Aug 2027	64,819	9,723	75,000	710,000
Sep 2027	63,327	9,499	75,000	785,000
Oct 2027	64,819	9,723	75,000	860,000
Year	726,230	108,935	860,000	860,000

Initial provisioning envelope

One time item	INR cap
OCR and initial document embeddings including tax allowance	1,000
Conditional sender registration or communication onboarding	6,000
Other initial vendor fees and provisioning headroom	3,000
Total initial envelope	10,000

At the assumed 10,000-page corpus and 50% OCR share, initial OCR and embeddings are approximately INR 805 before tax if OCR free allowances are ignored. The INR 1,000 cap covers that estimate. Sender registration and other provisioning amounts are conditional budget provisions, not quoted fees or authorised purchases. Domain cost is already included in recurring allocations and is not charged again here.

Calendar adjustment and funding margin

A simplified plan using twelve 30-day months would give INR 719,443 recurring cash need and INR 827,360 after 15% contingency. Actual calendar days increase those values to INR 726,230 and INR 835,165 respectively. The recommended allocation of INR 850,000 for recurring needs plus INR 10,000 for initial costs provides a further INR 14,835 above the calculated requirement.

6 Scenarios and spending sensitivities

Usage scenarios change the average daily active population while holding the registered base at 1,000, question mix constant, and non-generation envelopes unchanged. They are directional estimates; higher usage may require additional infrastructure capacity.

Scenario	DAU	Year questions	Recurring cash	Funding with reserve and setup
Low	100	164,250	474,448	555,615
Base	200	328,500	726,230	845,165
High	350	574,875	1,103,904	1,279,489

The base case supports the INR 8.60 lakh allocation. The high case requires approximately INR 12.79 lakh including contingency and the initial envelope. The scenarios do not predict how many registered users will become subscribers.

Sensitivity around the base case

Change	Annual recurring cash	Change from base
4 questions per DAU daily	670,279	-55,952
5 questions per DAU daily	782,182	55,952
25 percent higher generation cost	852,121	125,891

Every additional 1,000 questions costs approximately INR 1,405 in generation from January, or INR 1,658 including the provisional tax allowance. This excludes any capacity step-up. A 10% currency increase would add at most about INR 72,623 annually if every cost moved with USD; actual exposure should be lower because some allowances are domestic.

Conditional costs outside the core total

Payment processing depends on gross online collections. Use 3% of receipts including customer GST as a provisional collection-cost envelope, then replace it with the chosen gateway and recurring-mandate quote. For illustration, 150 subscriptions at INR 999 plus an assumed 18% customer GST produce INR 176,823 receipts and approximately INR 5,305 monthly collection costs. No revenue or gateway fee is embedded in the base cost plan. [8]

Managed point-in-time database recovery starts at USD 100 monthly, approximately INR 9,000 before tax, in addition to this base plan. A database upgrade from Small to Medium adds USD 45 monthly, or INR 4,050 before tax, assuming the same plan and credit. [3]

Paid legal-content licences, advanced document extraction, web-search grounding, voice, satellite imagery, enterprise SSO, contractual availability guarantees, dedicated or restricted hosting, security audits and procurement compliance services require a defined scope and supplier quote before being added. None is assumed for the existing text-document Q&A workload.

7 Budget controls and source register

Validation and review cadence

During the first month, compare metered provider usage with question volume and record model costs by simple, procedural and complex routes. Review active users, billable reasoning tokens, retries, OCR pages, database size, bandwidth and message counts. Use those measurements to replace the largest assumptions before the second monthly budget review.

Review the forecast if monthly question volume exceeds 30,000, average generation cost exceeds the applicable allowance by 20%, database disk reaches 80% of included capacity, or a required government procurement term changes the hosting or assurance scope. Set provider billing alerts and apply per-user usage allowances before offering broad commercial access.

Source register

Public pricing checked on 15 October 2026. Links identify the published rate or product reference used; estimates labelled as allowances remain internal planning assumptions.

[1] Google Gemini API pricing including text generation and embeddings

[2] Render maintained deployment example with service and queue costs

[3] Supabase pricing compute allowances backups and optional recovery

[4] Google Cloud Document AI OCR pricing

[5] Cloudflare R2 storage and operation pricing

[6] MSG91 WhatsApp platform subscription and message pricing reference

[7] TypeSafe AI published Jev input token rate

[8] Razorpay payment gateway pricing

[9] Cloudflare free plan services

[10] Project evidence: README.md describes the retrieval architecture; settings.py and services/gemini.py identify model defaults; generation_cost.py supplies the existing generation cost conventions; docs/cost_policy.md records selected evaluation runs; jev_settings.py and docs/jev.md describe optional Jev modes. These are repository references rather than live production measurements.

Decision basis

Approve an INR 860,000 first-year operating envelope for 1,000 registered users, subject to the stated activity, question-mix and standard-cloud assumptions. Treat the INR 10,000 initial allowance and optional integrations as controlled provisions. Development and maintenance labour remain outside this approval scope.